

BERKE TAYFUN AKSEKI

Data Analytics | Machine Learning | Backend Development

Rome, Italy • berketayfunakseki@gmail.com • berketayfunakseki.com • linkedin.com/in/berke-tayfun-akseki • github.com/berketayfunakseki

PROFILE

Computer Science and Management graduate from LUISS Guido Carli, Rome (Jul 2026). Shipped 7 end-to-end technical projects — an A/B-testing & difference-in-differences toolkit, analytical SQL work, machine learning APIs, RAG pipelines, a live job-market tracker, geo-analytics dashboards, a multiplayer game, and a portfolio site — all documented on GitHub. Combines Python / SQL engineering depth with business strategy training from a dual-track degree. Targeting junior roles in Data Analytics, ML Engineering, or Backend Development; open to relocation — Zurich, Basel, Zug or Milan.

EDUCATION

LUISS Guido Carli University — Rome, Italy

Sep 2023 – Jul 2026

Bachelor's Degree in Management and Computer Science (L-18)

- Thesis: "Tourism Dependence and Regional Macroeconomic Resilience in Italy" — difference-in-differences and event-study panel analysis across 21 NUTS 2 regions (2015–2024), Eurostat / ISTAT data, R; identified structural post-pandemic labour market hysteresis in high-tourism regions. Supervisor: Prof. Diletta Topazio.
- Top coursework (out of 30): Statistics 30, Finance & Financial Technologies 29, Microeconomics 27, Management of Technology 27, Algorithms 25, Data Analysis for Business 24.

TECHNICAL SKILLS

Languages & Frameworks: Python (Pandas, NumPy, scikit-learn), SQL, R, FastAPI, Pydantic, Django, C#, HTML / CSS, JavaScript

Data & Analytics: Power BI, KNIME, Streamlit, Excel, ETL, feature engineering, model evaluation, KPI dashboards, data visualisation

Tools & DevOps: Git / GitHub, Docker, GitHub Actions (CI/CD), PostgreSQL, SQLite, REST APIs, Unity 6, Blender, Jupyter, VS Code, RStudio

SELECTED PROJECTS

Portfolio Website — berketayfunakseki.com (Live) — HTML · CSS · JavaScript · GitHub Pages — berketayfunakseki.com

- Shipped a zero-dependency portfolio featuring an interactive JD-Match tool (recruiters paste a job description; client-side TF-IDF scores my profile live), a Recruiter / Engineering mode toggle with per-project architecture case studies, and visitor analytics — live at berketayfunakseki.com.

Causal Lab — A/B Testing & Difference-in-Differences Toolkit — Python · SciPy · statsmodels · FastAPI · Streamlit — github.com/berketayfunakseki/causal-lab

- Built a statistics-first toolkit for experimentation and hypothesis testing: two-proportion z-tests with confidence intervals, sample-size / power analysis, automated sample-ratio-mismatch detection, and a two-way fixed-effects difference-in-differences estimator with clustered standard errors — validated by tests that verify each method recovers a known, injected ground-truth effect from simulated data.

SQL Analytics Lab — Cohort Retention & Funnel Analysis — SQL (window functions, CTEs) · Python · Streamlit — github.com/berketayfunakseki/sql-analytics-lab

- Wrote production-style analytical SQL against a synthetic app-usage dataset: weekly cohort retention, step-over-step funnel conversion (LAG, FIRST_VALUE), and growth metrics (running totals, RANK vs DENSE_RANK) — surfaced a concrete product insight (steepest drop-off point in the funnel) directly from the data.

Labyscape — Multiplayer Co-op Horror Game (In Development, repo private) — Unity 6 · C# · Netcode for GameObjects · Blender

- Architected a 2–8 player co-op horror game: procedurally generated 18×18 DFS maze, host-authoritative Netcode multiplayer, NavMesh enemy AI, health / respawn systems, and a modular day/night cycle in Unity 6 URP — sole developer, targeting Steam.

Jobscope — Swiss / EU Job Market Tracker — FastAPI · SQLite · scikit-learn · Docker — github.com/berketayfunakseki/jobscope

- Automated pipeline aggregating 200+ live listings from RemoteOK and Adzuna APIs; deduplicates by normalised URL and scores each role 0–100 against a candidate profile via TF-IDF cosine similarity; JWT-authenticated FastAPI backend, containerised with Docker.

Regulatory RAG Assistant — FastAPI · Streamlit · FAISS · Docker — github.com/berketayfunakseki/regulatory-rag-assistant

- End-to-end Retrieval-Augmented Generation system over regulatory PDFs: document ingestion → chunking → embedding → FAISS vector search → FastAPI REST + Streamlit UI; containerised with Docker and GitHub Actions CI/CD.

Churn Prediction — ML API + Dashboard — scikit-learn · FastAPI · Streamlit

- Trained a customer-churn classifier with reproducible scikit-learn pipelines; served predictions via FastAPI endpoint; visualised model outputs in a Streamlit dashboard.

EXPERIENCE

LUISS Language Cafe — LUISS Guido Carli University

Feb 2026 – May 2026

Creative Manager Intern • Rome, Italy

- Owned end-to-end creative direction for the university language-exchange programme: designed social graphics, newsletters, and event copy distributed across LUISS channels.
- Launched multi-format content campaigns aligned with brand identity, increasing programme visibility and event attendance across student channels.

LANGUAGES

Turkish (native) • English (C1 — full professional) • French (B2) • Italian (A2)